

地理資料

國科會海洋學門資料庫

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海洋資料的…

- I 注意事項
- II 資料型態
- III 地圖投影
- IV 資料格式

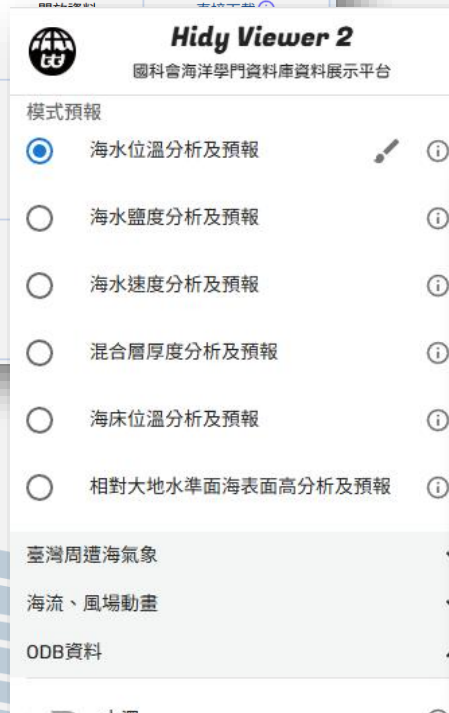
使用資料注意事項



- 資料生產者/提供者/處理者
- 如何生產(設備、技術、來源、後處理……)
- 資料解析度(時間、空間)
- 資料更新頻率
- 資料時間、時區
- 資料型式

來源	類別	資料集歸類	子項名稱	提供單位	說明	原始數據	資料集屬性	資料下載
觀測調查	船隻分布	航港局AIS	船舶自動識別(AIS)系統	交通部航港局	「船舶即時資訊」為一船舶自動識別系統(AIS)，預設每 60 秒更新一次船舶即時資訊，以13類子圖層區分不同船舶類型，透過顏色進行識別，除顯示船舶位置之外，個別……顯示更多	有	管制資料-智慧財產權資料①	來源連結① 前往圖臺
觀測調查	其他	能源署離岸風電	臺灣離岸風電潛力場址地理資訊	經濟部能源署	臺灣西部沿海36處離岸風電潛力場址之地理資訊。	有	開放資料	直接下載① 前往圖臺
觀測調查	其他	環境部離岸風電環評資料	離岸風電海域環評案件基本資料	環境部	離岸風電海域環評案件基本資料，專案代碼: 7b7244d0-a8a3-4b7c-9b90-3e1e5c7db71a、acead0……顯示更多	有		
觀測調查	生物	環境部離岸風電環評資料	離岸風電海域生態調查	環境部	離岸風電海域環評案件生態調查資料(搭配環評案件基本資料)，專案代碼: 7b7244d0-a8a3-4b7c-9b90-3e1e5c7db71a、<……顯示更多	有		
觀測調查	其他	環境部離岸風電環評資料	離岸風電海域空氣品質	環境部	離岸風電海域環評案件空氣品質調查資料(搭配環評案件基本資料)，專案代碼: 7b7244d0-a8a3-4b7c-9b90-3e1e5c7db71a……顯示更多	有		

NODASS資料頁面



Hidy資料說明

Classification

Full name	Global Ocean Physics Analysis and Forecast
Product ID	GLOBAL_ANALYSISFORECAST_PHY_001_024
Source	Numerical models
Spatial extent	Global Ocean · Lat -80° to 90° · Lon -180° to 179.92°
Spatial resolution	0.083° × 0.083°
Temporal extent	1 Nov 2020 to 4 Jul 2025
Temporal resolution	Hourly · Daily · Monthly
Elevation (depth) levels	50
Processing level	Level 4
Variables	Sea water potential temperature (T) · Sea surface temperature at sea floor (TOB) · Sea water salinity (S) · Sea water potential geoid (SSH) · Change in sea surface height due to change mean sea level (SSH) · Eastward sea water velocity (UV) · Stokes drift x velocity (UV, VSDXY) · Sea surface wave speed velocity (UV) · Surface sea water x velocity (UV) · Surface velocity due to tide (UV) · Surface sea water y velocity due to sigma theta (MLD) · Sea ice thickness (SIT) · Eastward sea ice velocity (SIUV) · Surface snow thickness (SNOW) · Age of sea ice temperature (IST) · Cell thickness · Change in sea floor loading · Global average sea level change due to change in sea level · Model level number at sea floor · Ocean dynamic sea level · Sea binary mask · Sea floor depth below geoid · Sea ice area fraction · Sea ice speed · Sea water pressure at sea floor
Feature type	Grid
Blue markets	Polar Environment Monitoring · Policy & Governance · Education · Public Health & Recreation · Science & Innovation · Extremes & Hazards & Safety · Coastal Services · Natural Resources & Energy · Trade & Marine Navigation
Projection	Equirectangular
Data assimilation	Sea Level · Sea Ice Concentration and/or Thickness · SST · In-Situ TS Profiles
Update frequency	Daily – 12:00 UTC; 15th day of month at 12:00 UTC · Monthly
Format	NetCDF-3 · NetCDF-4
Originating centre	Mercator Océan International
Last metadata update	26 November 2024

Description

Notifications

Data access

Contact

DOCUMENTATION

User Manual

Quality Information Document

Synthesis Quality Overview

Product Quality Dashboard

Roadmap

Licence

How to cite

DOI

10.48670/moi-00016

Overview

The Operational Mercator global ocean analysis and forecast system at 1/12 degree is providing 10 days of 3D global ocean forecasts updated daily. The time series is aggregated in time in order to reach a two full year's time series sliding window.

This product includes daily and monthly mean files of temperature, salinity, currents, sea level, mixed layer depth and ice parameters from the top to the bottom over the global ocean. It also includes hourly mean surface fields for sea level height, temperature and currents. The global ocean output files are displayed with a 1/12 degree horizontal resolution with regular longitude/latitude equirectangular projection.

50 vertical levels are ranging from 0 to 5500 meters.

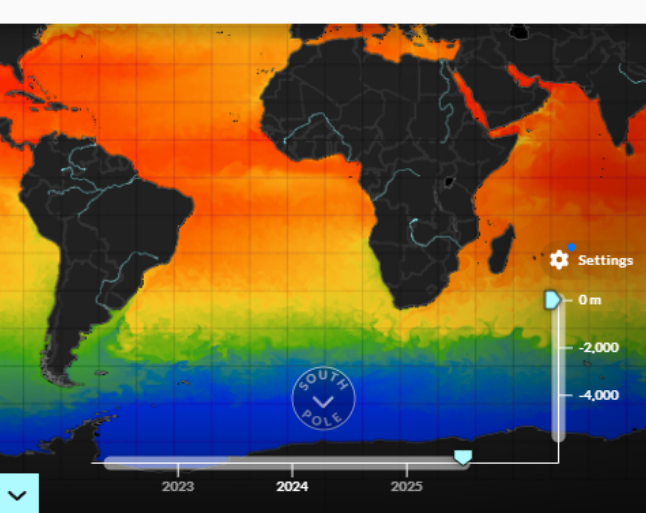
This product also delivers a special dataset for surface current which also includes wave and tidal drift called SMOC (Surface merged Ocean Current).

DOI (product):

<https://doi.org/10.48670/moi-00016>

Sea water potential temperature

2025/07/02 · -0.5 m · Global daily



Explore in MyOcean Pro

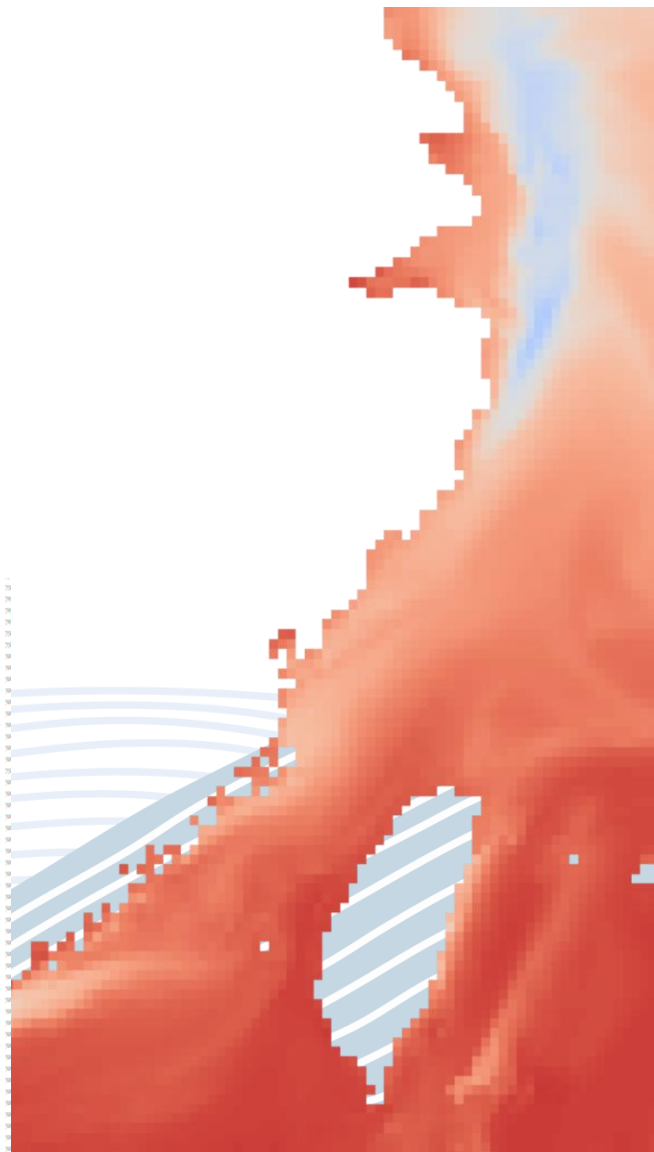
CMEMS資料頁面

資料型態



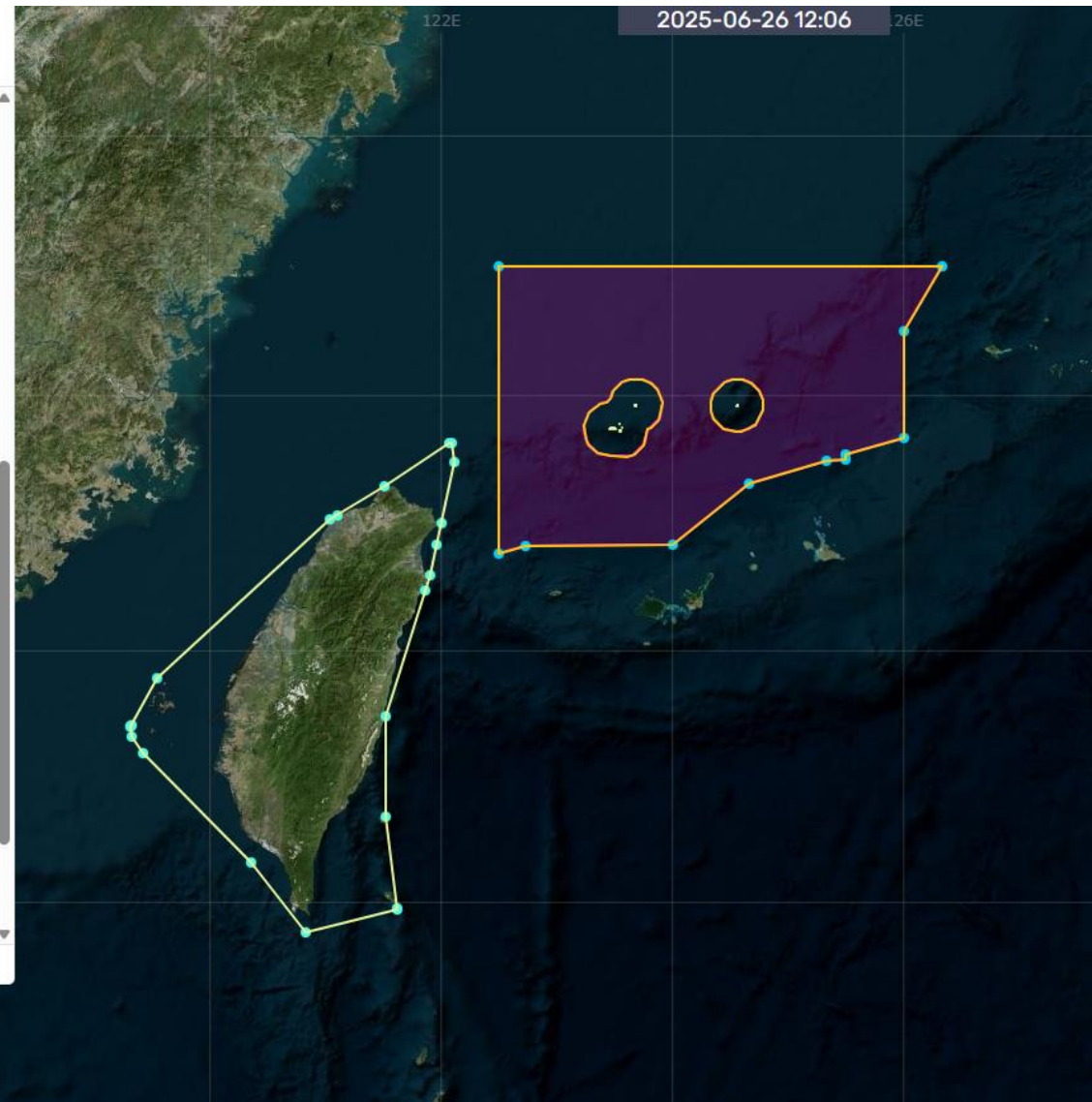
- 資料來源
 - 衛星資料(大範圍、固定周期)
 - 雷達/聲納/航空資料(小範圍)
 - 儀器資料(點)
- 資料處理
 - 觀測資料 → 珍貴不可逆!!
 - 推估/模式資料 → 好用但須注意處理方法
- 資料型式
 - 網格資料
 - 向量資料
 - 文字、數字、表格、影像…等

- [illegible]

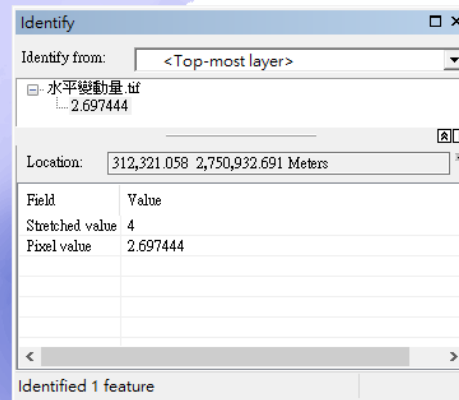




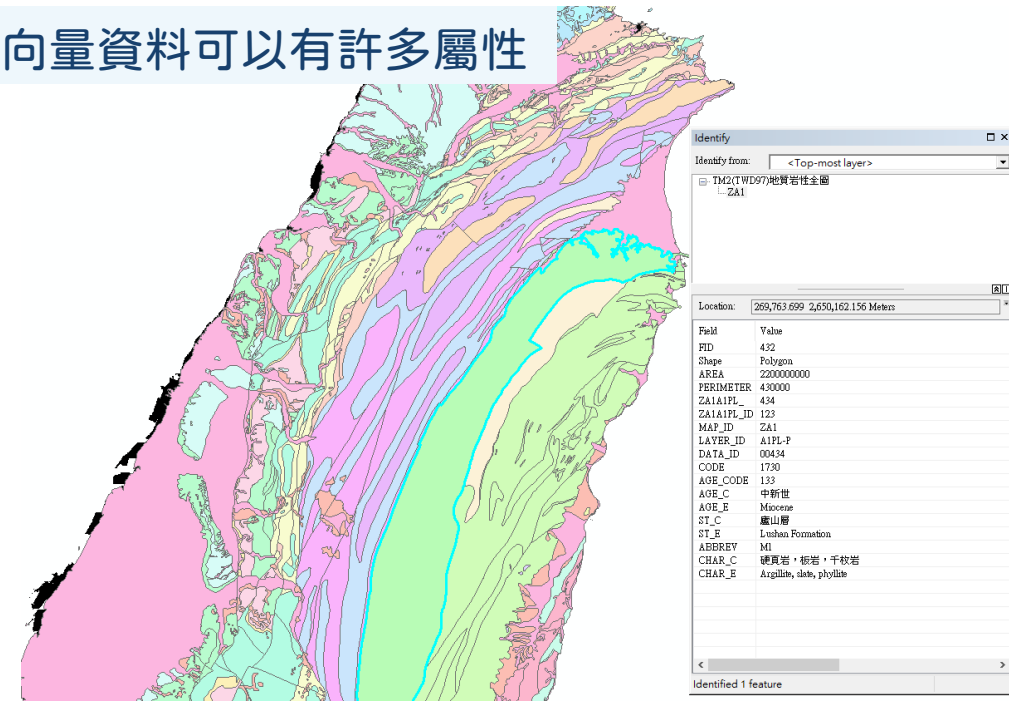
- 向量資料(vector data)
 - 點→線→面
 - 可以無限放大
 - 可以附加許多屬性資料
 - 點位過多(形狀不規則)可能需要考慮簡化縮小檔案
 - eg., .shp, .json, .kml...



網格僅有pixel value一個屬性(stretched value為繪圖的顏色，非資料本身)



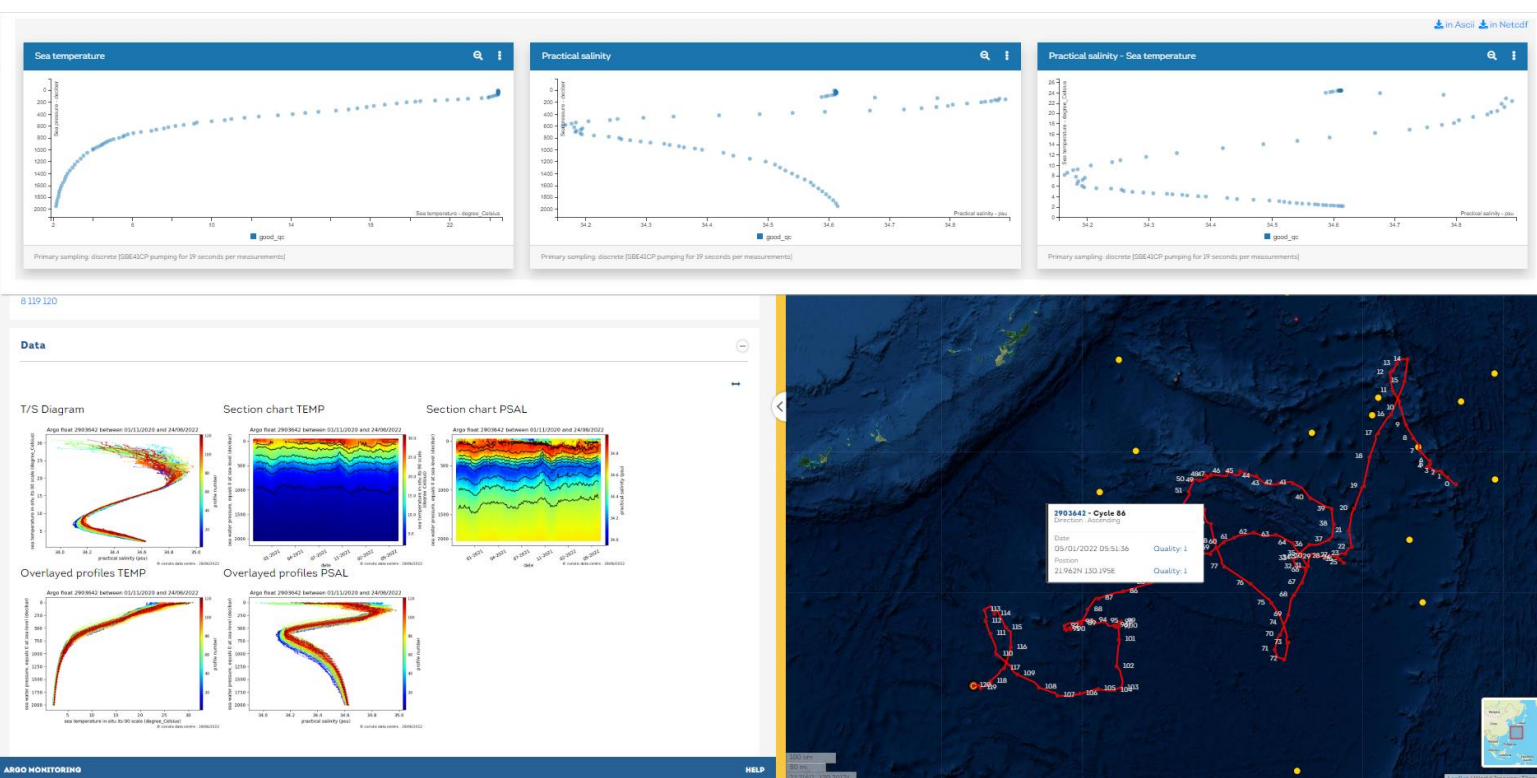
向量資料可以有許多屬性



網格

向量

- 儀器、向量、觀測



Argo浮標資料

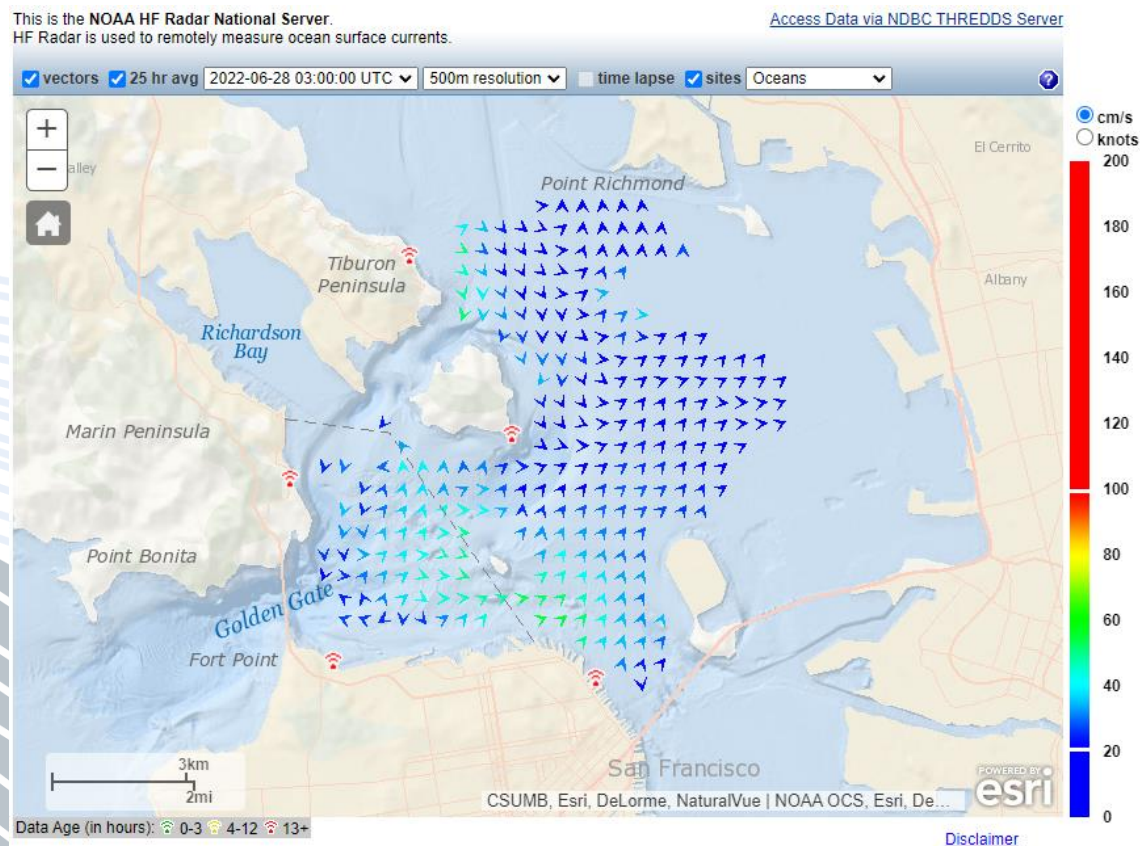


Hidy 中央氣象署海象觀測

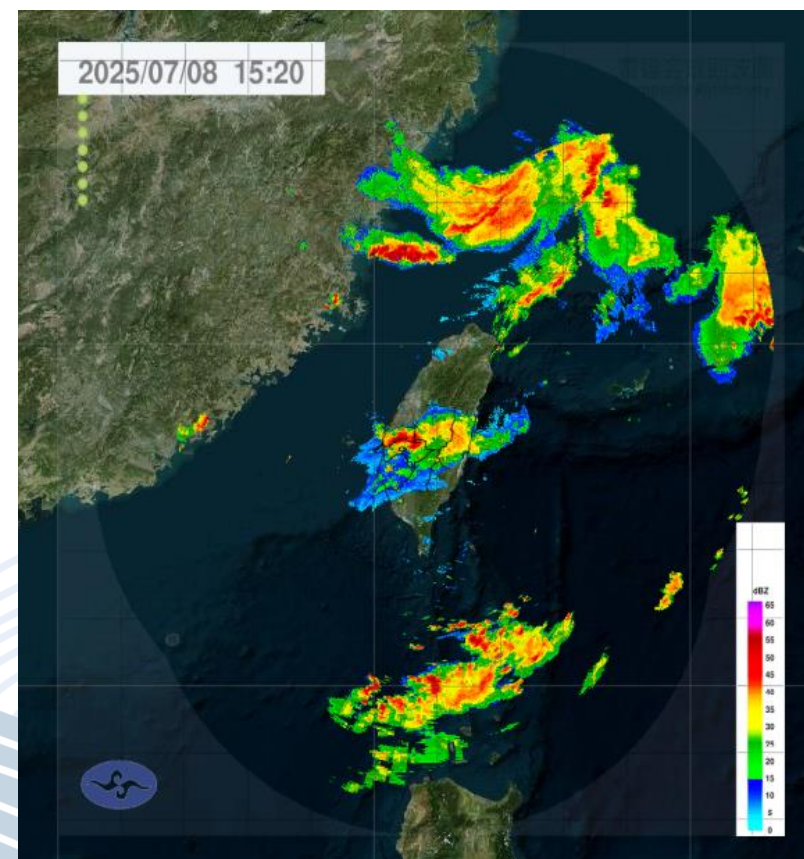
<https://fleetmonitoring.euro-argo.eu/dashboard>



- 雷達、網格、觀測/推估



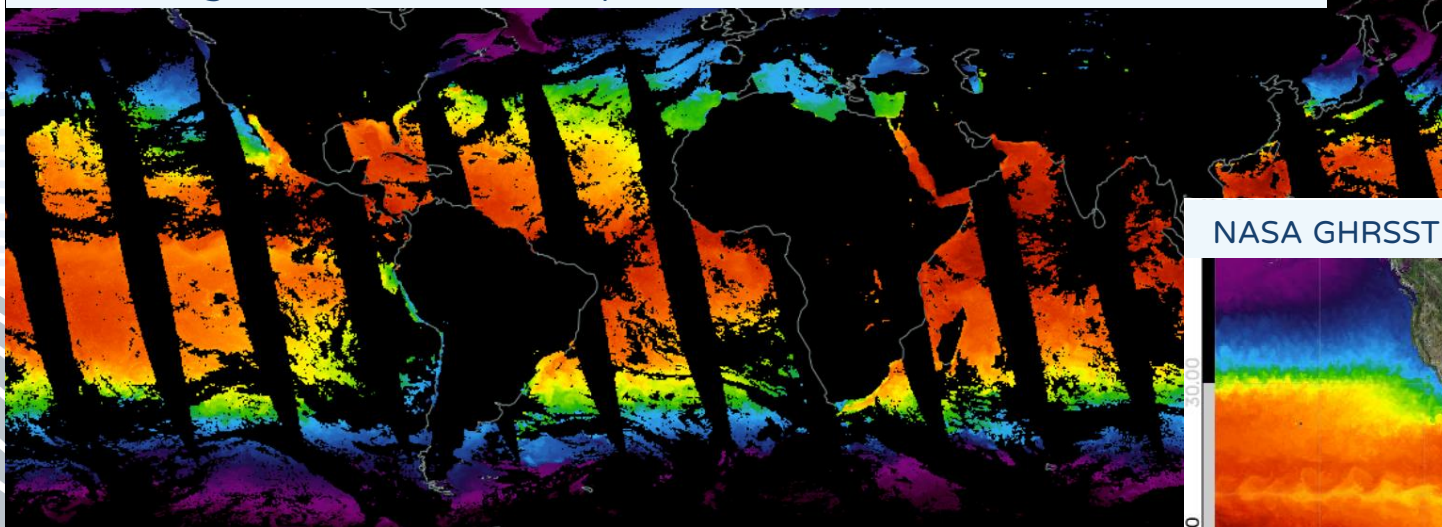
NOAA HF Radar

<https://hfradar.ndbc.noaa.gov/>

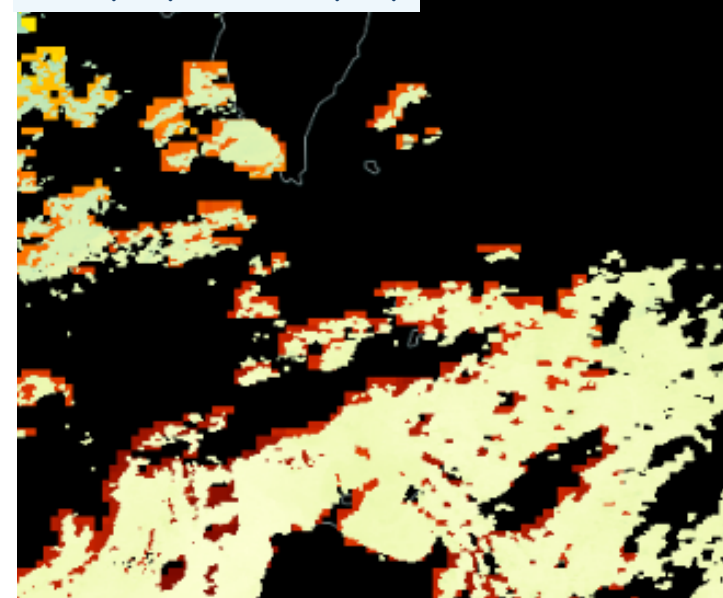
Hidy 中央氣象署雷達回波

- NASA資料處理等級 L0 ~ L4
 - L3網格化
 - L4包含了模式及其他計算

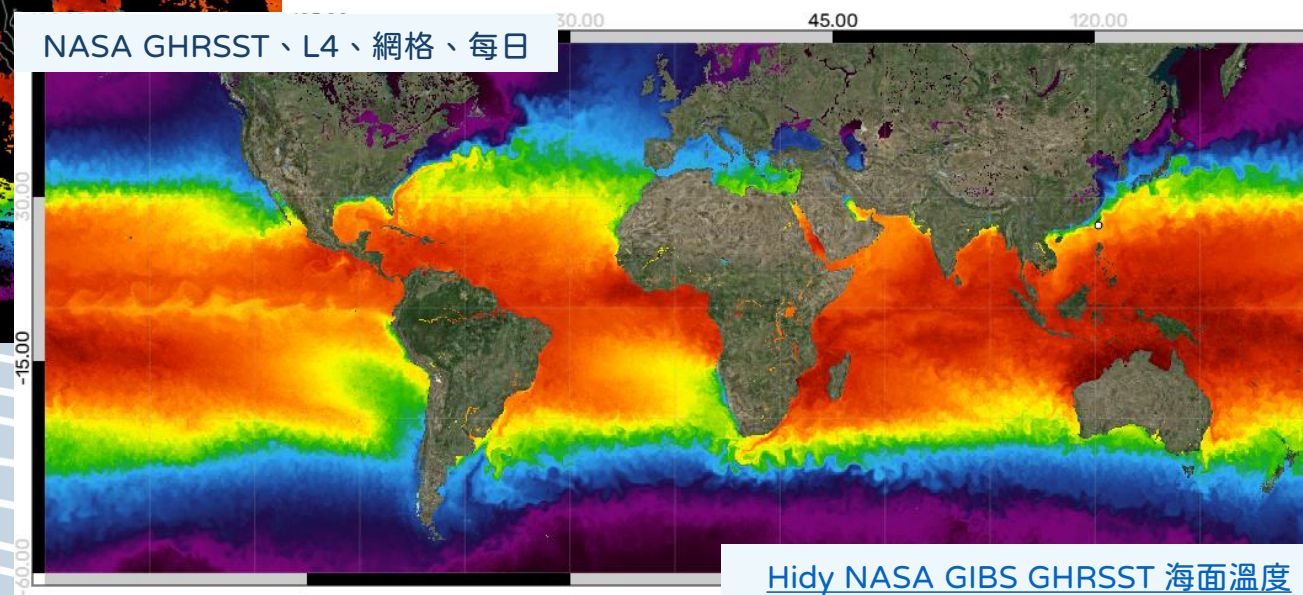
L3 Along-track, 9km, Daily, Terra/MODIS, EOSDIS NASA



L2 (前) vs L3 (後)



NASA GHRSSST、L4、網格、每日



Hidy NASA GIBS GHRSSST 海面溫度

[CMEM範例: Hidy 海水位溫分析預報](#)

地圖投影



- 1 基本概念
- 2 投影及座標
- 3 高程

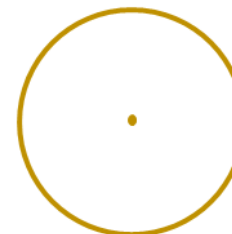


- 真實地表
 - 人類發展各種方法描述、簡化真實世界
- 大地水準面(Geoid)：
 - 又稱平均海平面 (Mean Sea Level) ，
 - 只考慮地球重力和自轉的海水面在海水靜止狀況下與重力的等位面
 - 由重力資料計算之模型
- 參考橢球體(Reference ellipsoid)：
 - 「完美」平滑的數學橢球模型
 - 例：GRS80、WGS84橢球

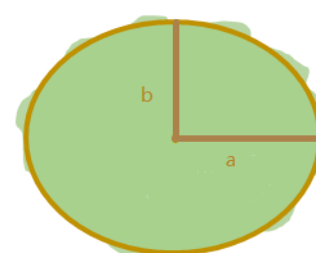
Planet earth



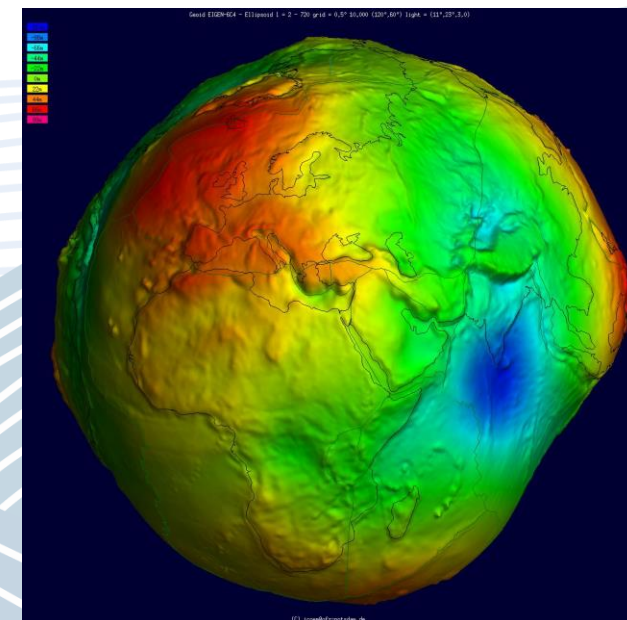
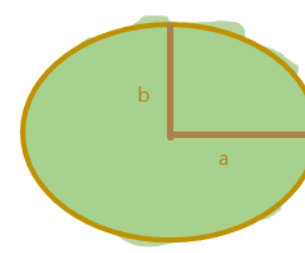
Spheres



WGS84

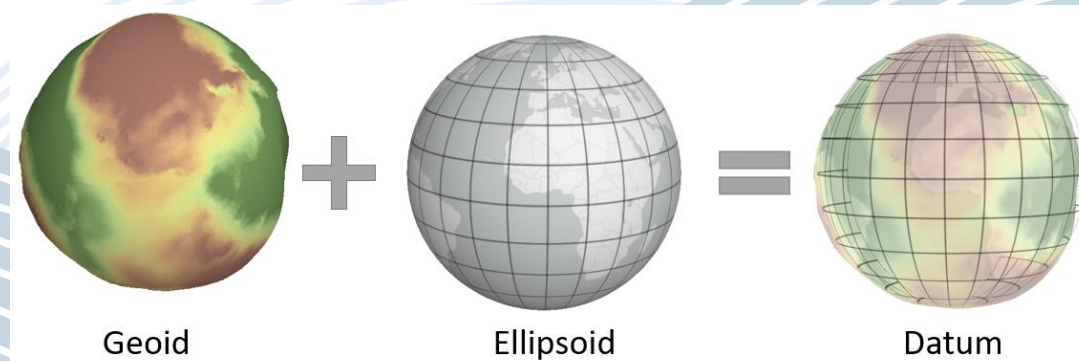
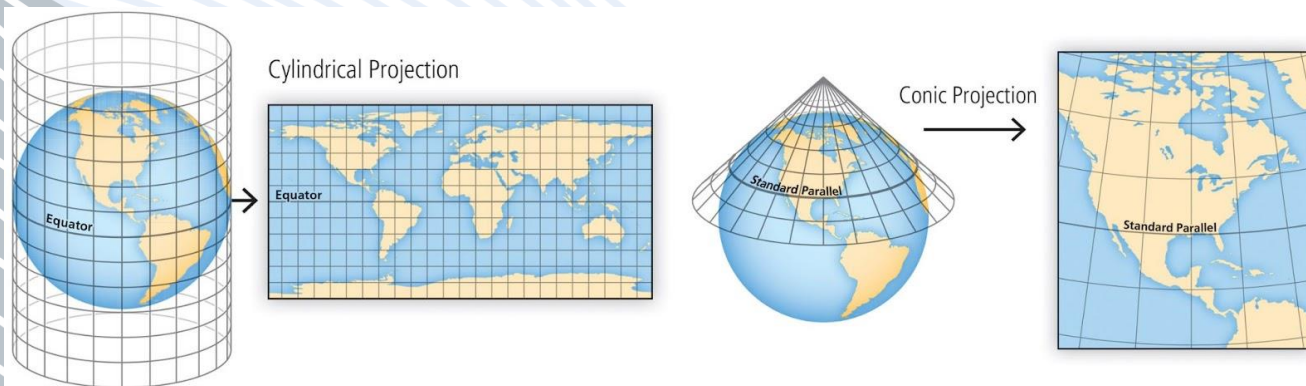
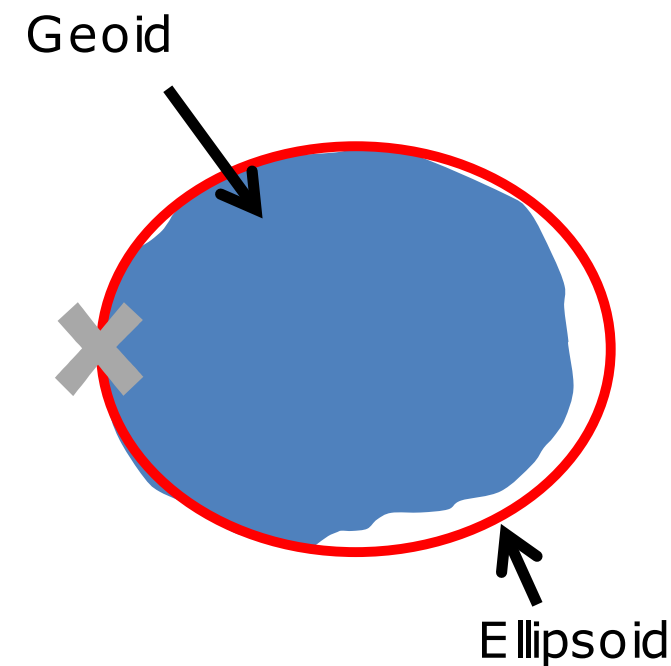


GRS80

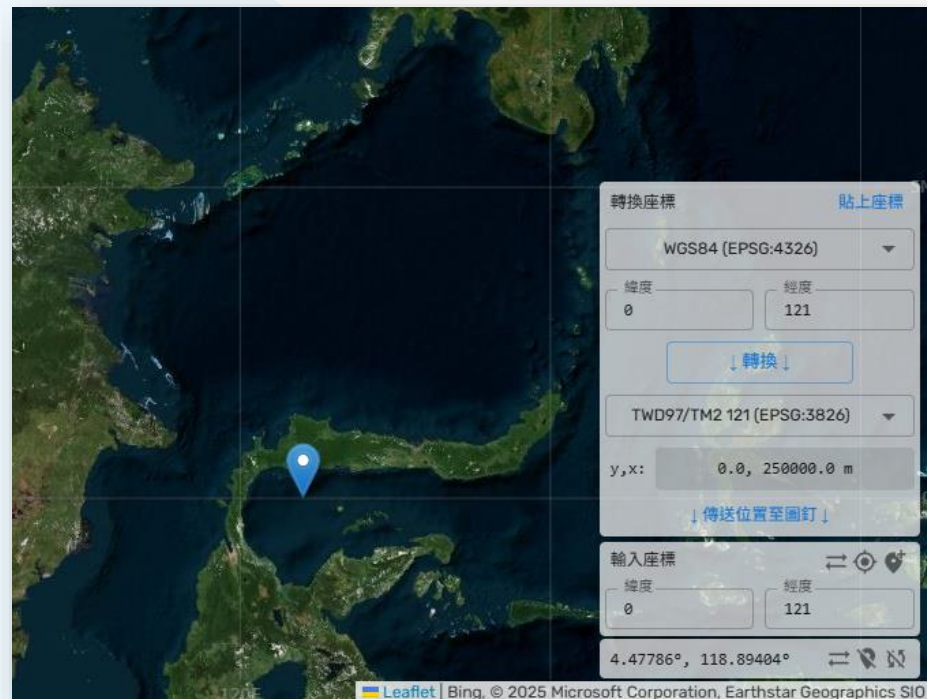


- <http://doi.org/10.5194/essd-11-647-2019>
- <https://support.virtual-surveyor.com/en/support/solutions/articles/1000261329-what-is-an-ellipsoid->

- 大地(測量)基準(datum) :
 - 橢球體衍生之經緯座標系統，盡可能貼合地表
 - 例：WGS84、NAD83、TWD97
- 投影(projection) :
 - 將曲面轉換為平面，會有某部分犧牲
 - 例：Mercator、Mollweide、TWD97 TM2



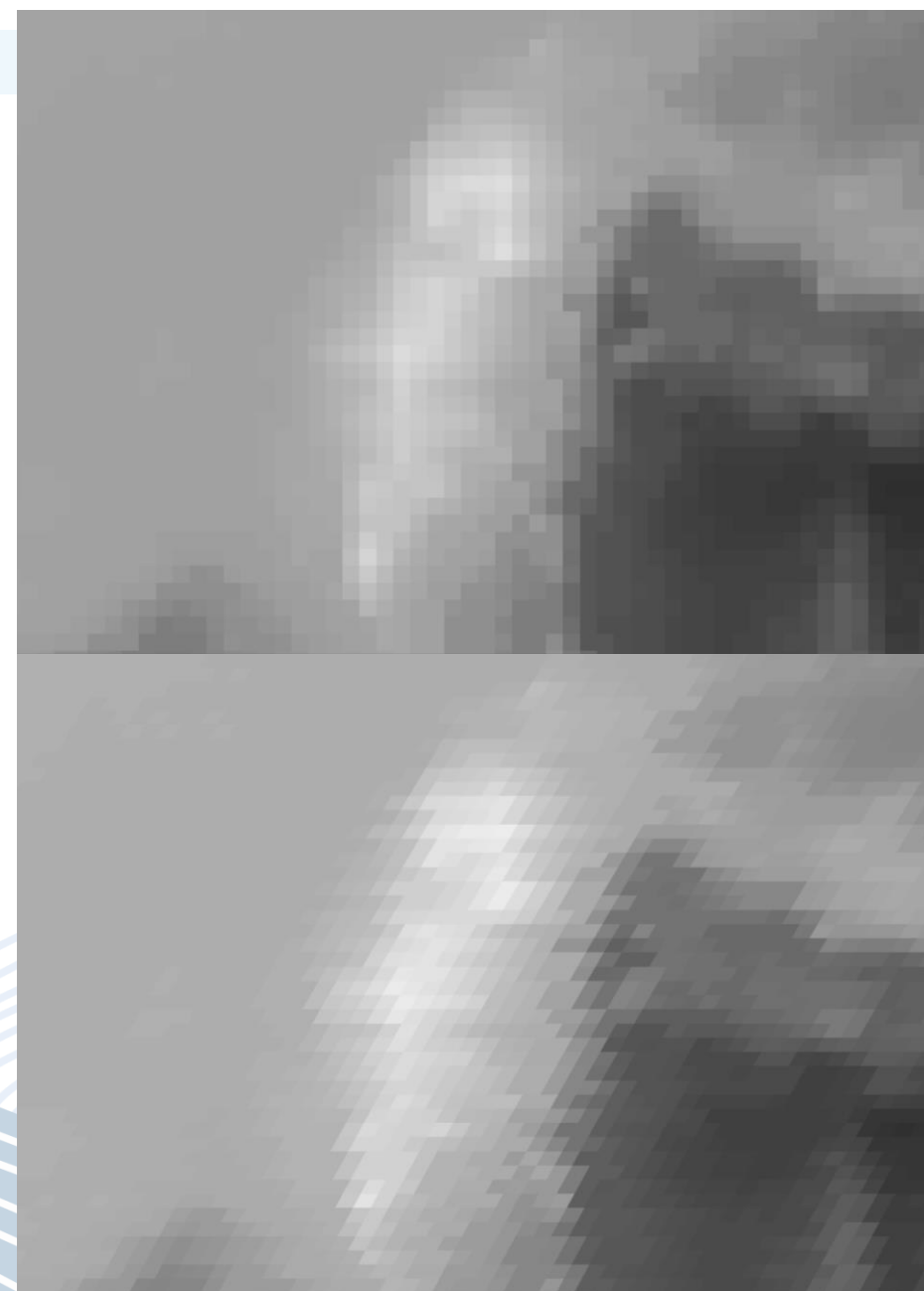
- 未投影前座標通常是經緯度，只能估算距離、面積
 - Raster可以用網格估算面積及長度，但精度不高
- 投影後可以直接計算距離、面積。不同投影法會得到不同的結果
- 投影後座標通常會是長度單位(但也可能是經緯度)
 - EPSG:4326: WGS84經緯度座標系統
 - WGS84 datum, WGS84橢球體
 - (121°32'14", 25°1'16")
 - EPSG:3857: Web Mercator投影，原點為(0°, 0°)
 - WGS84 datum, WGS84橢球體
 - (13529461.69, 2878337.87)
 - 0度經線以東13,529,461.69公尺
 - 赤道以北2,878,337.87公尺
 - EPSG:3826(TWD97 TM2): 121E為中心二度分帶投影並向西移250,000公尺為原點
 - TWD97 datum, GRS80橢球體
 - (304218.45, 2768223.38)
 - 121度以東54,218.45公尺 (304,218.45 - 250,000 = 54,218.45)
 - 赤道以北2,768,223.38公尺
- 使用WMS時，bounding box(bbox)參數須注意CRS單位及原點

[Hidy座標及投影轉換小工具](#)



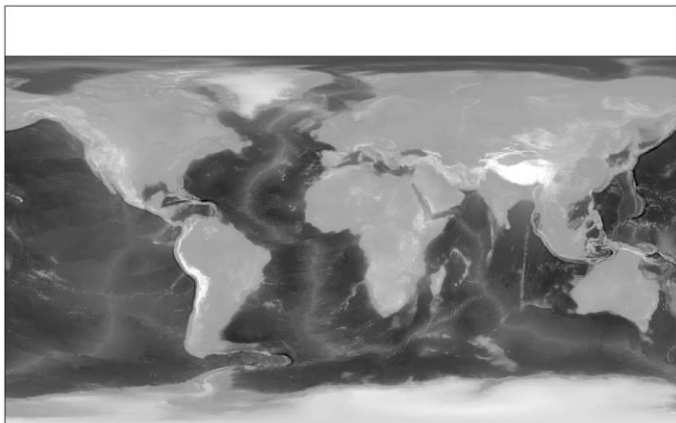
在EPSG:3857底圖疊上EPSG:3857(左)及EPSG:4326(右)地圖

- 中央氣象局 海流模式-臺灣海域海流預報數值模式資料-海流數值模式預報資料
- World digital elevation model (ETOPO5)

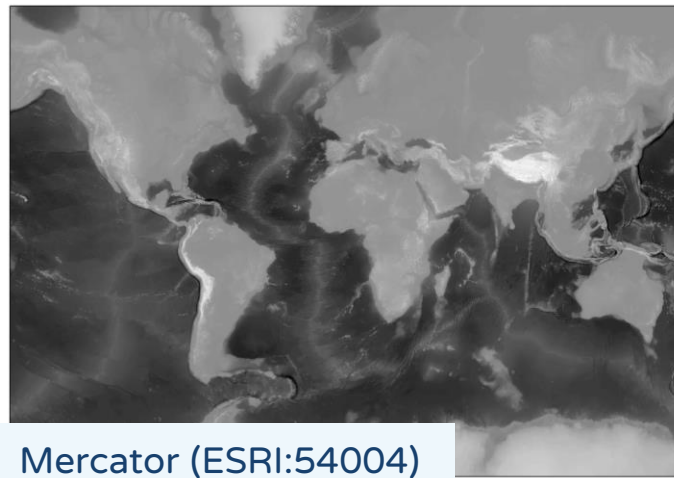


EPSG:4326坐標系中的EPSG:4326(上) vs Mollweide投影(下)

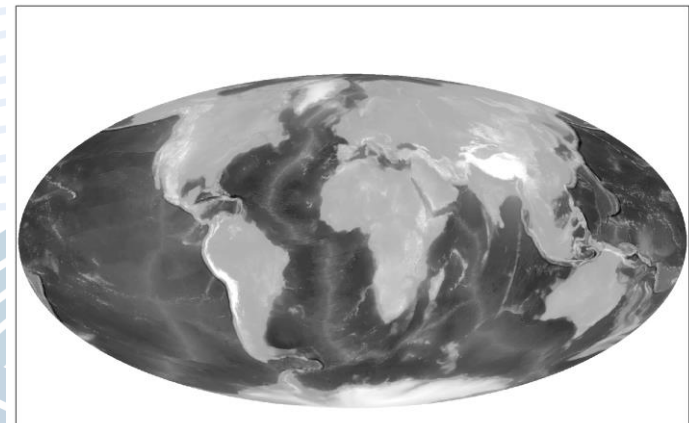
- 所有的地圖產品都需要注意座標系統及投影方法
- Spatial reference system(SRS) 或 Coordinate reference system (CRS)
 - 參考橢球體(Reference Ellipsoid)
 - 大地基準(datum)
 - 原點(Origin)
 - 單位
 - *投影方式(Projection)



WGS84 (EPSG:4326)



Mercator (ESRI:54004)



Mollweide (ESRI:54009)

- EPSG編號系統 EPSG:4326：WGS84座標系統(GPS使用)
 - EPSG:3857 (900913)：Web Mercator平面投影
 - EPSG:3826：TWD97/TM2 121 (TWD97二度分帶，以121度為中心)
- 轉換座標、查詢地圖位置及EPSG詳細資料 <https://epsg.io/>

EPSG:3821

TWD67

Attributes

Unit: degree (supplier to define representation)

Geodetic CRS: TWD67

Datum: Taiwan Datum 1967

Ellipsoid: GRS 1967 Modified

Prime meridian: Greenwich

Data source: EPSG

Information source: National Land Surveying and Mapping Center (NLSC), <http://www.nlsc.gov.tw>

Revision date: 2008-08-11

Scope: Geodesy.**Remarks:** Shares the same origin point with the earlier Hu Tzu Shan system (CRS code 4236) but away from this point coordinates differ. Do not confuse!**Area of use:** Taiwan, Republic of China - onshore - Taiwan Island, Penghu (Pescadores) Islands.**Coordinate system:** Ellipsoidal 2D CS. Axes: latitude, longitude. Orientations: north, east. UoM: degree

Covered area powered by MapTiler

**Center coordinates**

120.66 23.61

WGS84 bounds:

119.25 21.87

122.06 25.34

Taiwan, Republic of China - onshore - Taiwan Island, Penghu (Pescadores) Islands.

EPSG:3828

TWD67 / TM2 zone 121

Attributes

Unit: metre

Geodetic CRS: TWD67

Datum: Taiwan Datum 1967

Ellipsoid: GRS 1967 Modified

Prime meridian: Greenwich

Data source: EPSG

Information source: National Land Surveying and Mapping Center (NLSC), <http://www.nlsc.gov.tw>

Revision date: 2012-02-13

Scope: Cadastre, engineering survey, topographic mapping.**Remarks:** Except for cadastral use, replaced by TWD97 / TM2 zone 121 (CRS code 3826).**Area of use:** Taiwan, Republic of China - onshore - Taiwan Island.**Coordinate system:** Cartesian 2D CS. Axes: easting, northing (X,Y). Orientations: east, north. UoM: m.

Covered area powered by MapTiler

**Center coordinates**

252551.25 2611288.37

Projected bounds:

145616.57 2419172.28

359551.35 2803869.61

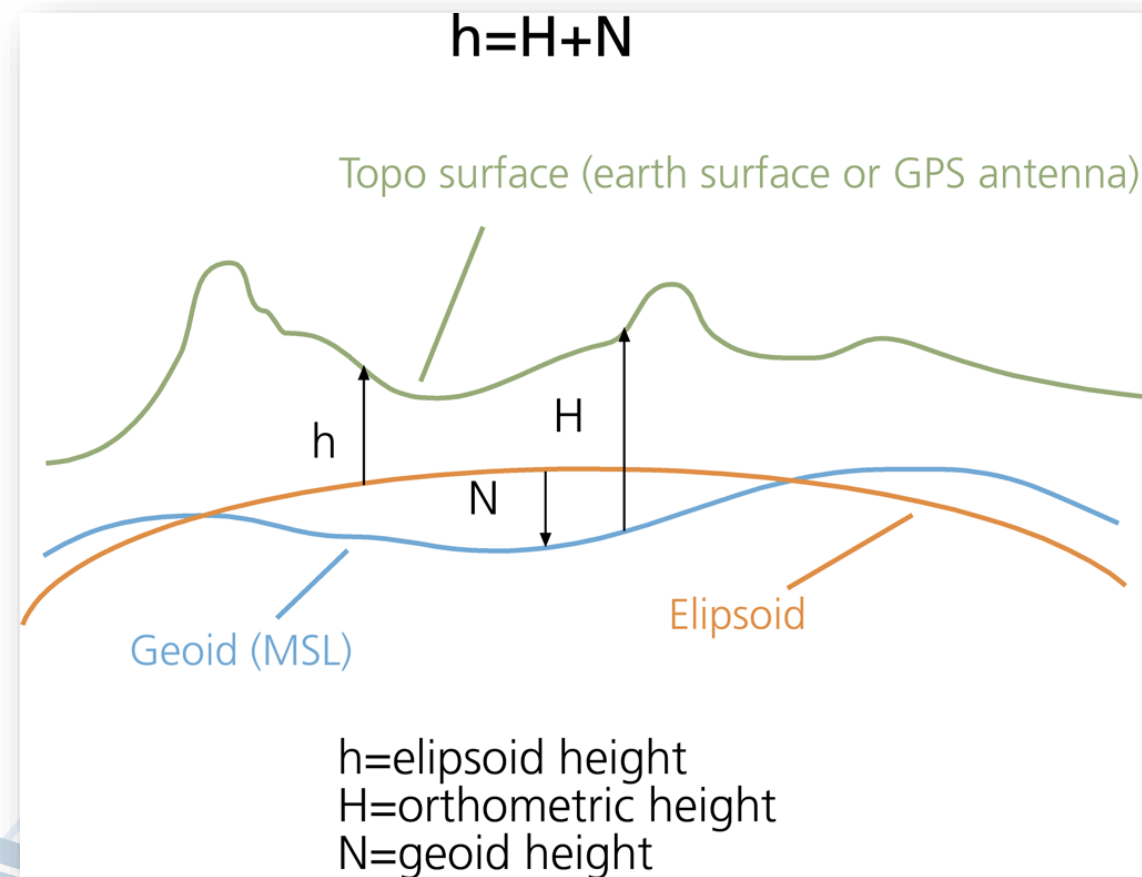
WGS84 bounds:

119.99 21.87

122.06 25.34

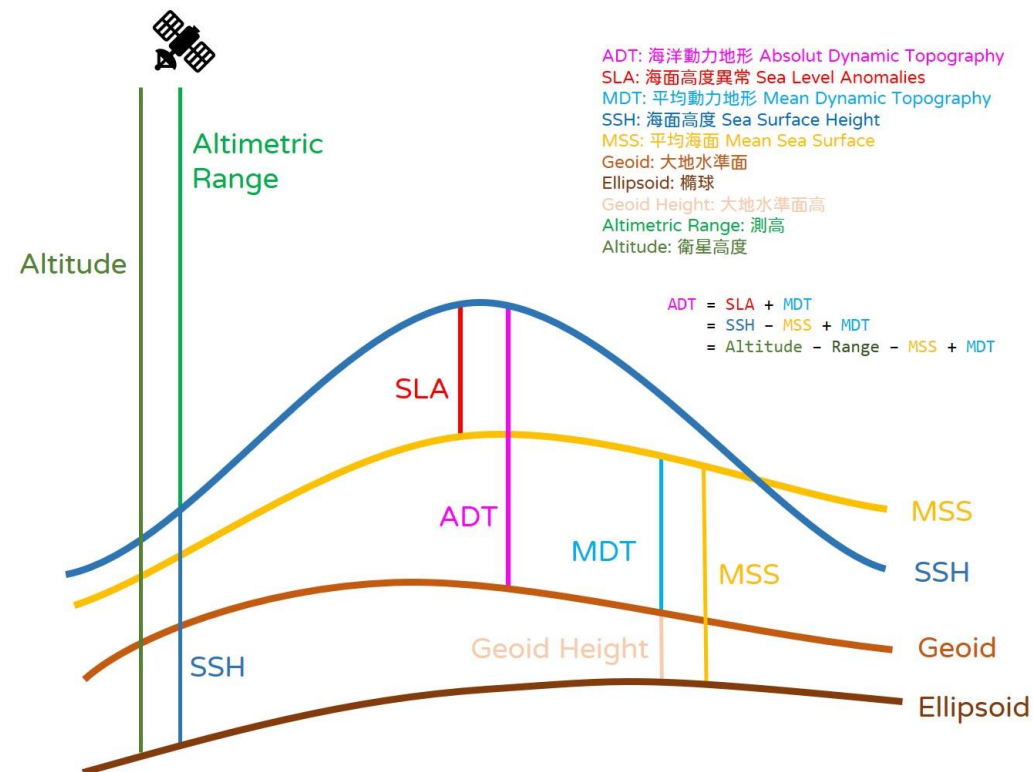
Taiwan, Republic of China - onshore - Taiwan Island.

- 橢球高度(h)以橢球體為基準
- 大地起伏(N)為橢球體與大地水準差
- 正高(H)= $h - N$
 - 常用之“海拔高度”
 - 無法直接測量
 - 測量點與基準點再比較高度
- *GPS將與橢球體同原點之地心地固坐標系(ECEF，一種笛卡兒坐標系)座標轉換為大地座標(經緯度及橢球高 h)

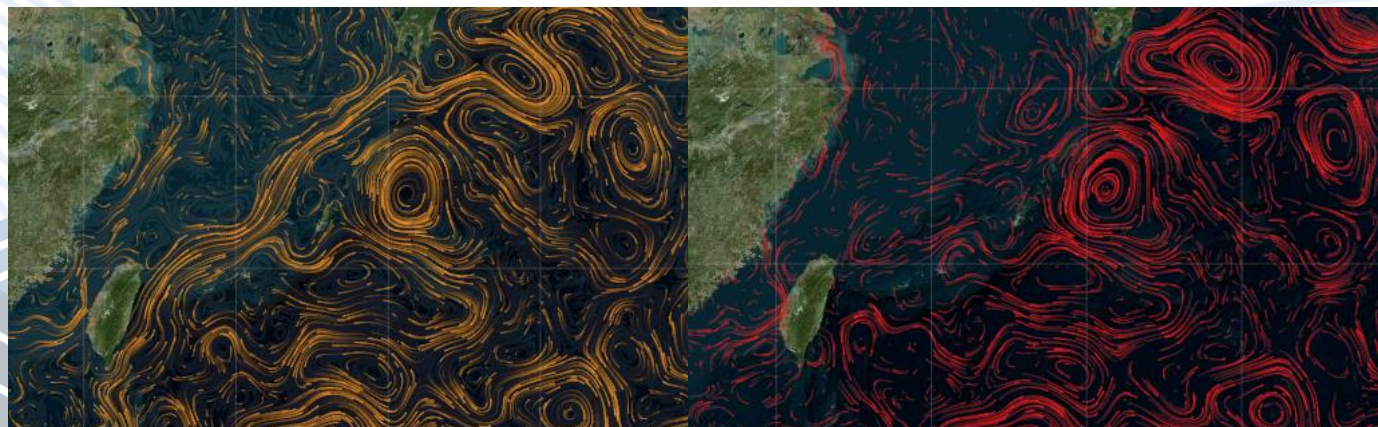


<https://www.esri.com/news/arcuser/0703/geoid1of3.html>

- AVISO衛星高度資料
 - MSS: 測量之海水面與參考橢球體之長期(至少>1年)平均高差
 - MSS平均掉潮汐、風等週期或短期影響
 - SLA適用於研究短期潮汐、波浪…等
 - MDT:MSS與大地水準面之高差
 - 主要受重力和地球自轉外的海水質量分布、溫度、鹽度等影響
 - ADT適合研究洋流循環與氣候變遷



Hidy地轉流速(geoid)



Hidy ADT(左) vs SLA(右)計算之海流

Hidy地轉流速(MSS)



潮汐基準面

- TWVD2001: 參考基隆平均海水面設立之臺灣水準原點為準
- TWCD2021: 參考33個長期潮位站建立之深度模型
- NTDE: 美國使用19年(1983-2001)平均海水面

*Hidy潮汐估計以TPXO9-atlas-v5模式計算，基準為NTDE之平均海水面MSL

- 潮汐預報提供3種不同基準面起算的潮位值，因使用者的用途差異，慣用的基準面不同(例如：工程、漁釣、航海)。
 - 相對臺灣高程系統 (本島為臺灣高程基準 TWVD2001，以基隆港平均海水面為0；離島為當地水準高程)
 - 相對當地平均海平面 (TWCD 2021，以當地平均海平面為0)
 - 相對海圖 (TWCD 2021，以當地最低低潮位為0)

CWA潮汐預報

日期	潮汐	時間	相對臺灣 高程系統	相對當地 平均海平面	相對海圖
新北市 貢寮 潮差：小 未來30天	滿潮 	00:27	40	30	134
	乾潮 	07:59	14	5	108
	滿潮 	12:29	24	15	118
	乾潮 	18:06	14	5	108

資料格式



1 資料時間

2 常見格式

給人看的(human readable)

- ISO8601格式
 - YYYY-MM-DDThh:mm:ssZ → UTC
 - YYYY-MM-DDThh:mm:ss+00:00 → UTC
 - 2025-07-17T10:00:00+08:00 → UTC+8
 - 2025-07-17T10:00:00 → 未知時區

方便計算的(累積)

- POSIX/Unix time
 - UTC 1970/1/1 00:00:00起累積秒數，連續無潤秒
- GPS時間
 - UTC 1980/1/6 00:00:00起累積秒數，連續無潤秒
- Excel
 - 1900/1/1起累積日數(Mac版1904年)，日以下以小數點表示
 - 無法顯示潤秒 (23:59:60)

- 國際原子時(TAI)
 - UT2 1958/1/1 00:00:00起，連續、無潤秒
 - 原子鐘時間
- 世界協調時間(UTC)
 - 不連續，以TAI為基礎加上潤秒
 - (應該是)目前在地球上最常使用的時間標準
- 換算
 - $\text{TAI-GPS}=19\text{秒(恆定)}$
 - $\text{TAI-UTC}=37\text{秒}$ ； $\text{GPS-UTC}=18\text{秒}$ (自2017)

<https://www.epochconverter.com/>

- CSV (Comma-Separated Values)
 - 逗號分隔
 - 優: 簡單，歷史悠久
 - 缺: 遇到資料內有逗號(如外文名、地址)較為麻煩
- TSV (Tab-Separated Values)
 - Tab分隔
 - 優: 分隔符號(\t)比逗號少見
 - 缺: 檢視資料時可能tab或空白分不清楚
- 但對pandas來說處理難度差別不大
 - `df_csv = pd.read_csv("data.csv")`
 - `df_tsv = pd.read_csv("data.tsv", sep='\t')`

```
{
  "全名": "JavaScript Object Notation",
  "使用情境": "網頁資料經常使用",
  "形式": "以大括弧包裹key:value，以半形逗號分隔大括弧",
  "接受巢狀": true,
  "geojson": {
    "description": "JSON的一種，包含type和地理座標 (WGS84)，藉此描繪出圖形",
    "必要條件": "Feature必須要有geometry，且包含coordinates才會是GeoJSON。很多個feature object可以組成一個features陣列，包含在FeatureCollection內",
    "type": ["Point", "LineString", "Polygon", "MultiPoint", "MultiLineString", "MultiPolygon"]
  }
}
```

- Python: json

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "id": 0,
      "geometry": {
        "type": "Point",
        "coordinates": [
          126.0,
          29.3
        ]
      },
      "properties": {
        "FID": 0,
        "Name": "1",
        "lat": "29°18'00\"N",
        "lng": "126°00'00\"E",
        "Y": 29.3,
        "X": 126
      }
    },
    {
      "type": "Feature",
      "id": 1,
      "geometry": {
        "type": "Point",
        "coordinates": [
          126.0,
          25.666667
        ]
      },
      "properties": {
        "FID": 1,
        "Name": "2",
        "lat": "25°40'00\"N",
        "lng": "126°00'00\"E",
        "Y": 25.66666667,
        "X": 126
      }
    }
  ]
}
```

GeoJSON Point

- ```
<?xml version="1.0"?>
<title>Extensible Markup Language</title>
<使用>HTML、GPX、KML、MS Office...等</使用>
<巢狀>可以，且通常會有</巢狀>
<格式>
 <包裝方式>以tag包裹資料</包裝方式>
 <屬性 style="color:red">tag可有屬性</屬性>
</格式>
```
- Python: xml.etree.ElementTree, html

```
<dataset>
 <datasetInfo>
 <datasetDescription>海象預報模式資料(資料，範圍經度110-126度，緯度7-36度)</datasetDescription>
 <parameterSet>
 <parameter>
 <parameterName>緯度格點數</parameterName>
 <parameterValue>291</parameterValue>
 </parameter>
 </parameterSet>
 </datasetInfo>
 <time>
 <datetime>2022-06-12T00:00+00:00</datetime>
 </time>
 <location>
 <lat> 7.0</lat>
 <lon>110.0</lon>
 <weatherElement>
 <elementName>海表溫度</elementName>
 <elementValue>
 <value>nan</value>
 <measures>度</measures>
 </elementValue>
 </weatherElement>
 <weatherElement>
 <elementName>橫向流速</elementName>
 <elementValue>
 <value>nan</value>
 <measures>m/s</measures>
 </elementValue>
 </weatherElement>
 </location>
</dataset>
```

CWA海象預報.xml資料

```
<?xml version='1.0' encoding='UTF-8'?>
<kml xmlns="http://www.opengis.net/kml/2.2" xmlns:gx="http://www.google.com/kml
xsi:schemaLocation="http://www.opengis.net/kml/2.2 http://schemas.opengis.net/k
http://code.google.com/apis/kml/schema/kml22gx.xsd">
 <Document id="空版臺灣海峽中線">
 <name>空版臺灣海峽中線</name>
 <snippet/>
 <description><![CDATA[<DIV
STYLE="text-align:Left;"><DIV><DIV><P>由國防部定義之臺灣海峽中線。本圖
用任何政治、經濟、法律、導航及其他用途。國科會海洋學門資料庫對於本圖資的可能
/></P></DIV></DIV></DIV>]]></description>
 <Folder id="FeatureLayer0">
 <name>空版臺灣海峽中線</name>
 <snippet/>
 <description><![CDATA[<DIV
STYLE="text-align:Left;"><DIV><DIV><P>由國防部定義之臺灣海峽中線。本圖
適用任何政治、經濟、法律、導航及其他用途。國科會海洋學門資料庫對於本圖資的可
/></P></DIV></DIV></DIV>]]></description>
 <Placemark id="ID 00000">
 <name>空版臺灣海峽中線</name>
 <snippet/>
 <description><![CDATA[<html xmlns:fo="http://www.w3.org/1999/XSL/Format"
http-equiv="Content-Type" content="text/html"><meta http-equiv="content-t
0px;overflow:auto;background:#FFFFFF;"><table
style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;wid
style="text-align:center;font-weight:bold;background:#9CBCE2"><td>空版臺灣
style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;wid
3px"><tr><td>FID</td><td>0</td></tr><tr bgcolor="#D4E4F3"><td>name</td><t
type="text/javascript">function changeImage(attElement, nam
attElement; document.getElementById('imageName').innerHTML
<styleUrl>#LineStyle00</styleUrl>
 <MultiGeometry>
 <LineString>
 <extrude>0</extrude>
 <tessellate>1</tessellate><altitudeMode>clampToGround</altitudeMode>
 <coordinates>118.0,23.0,0.0 122.0,27.0,0.0</coordinates>
 </LineString>
 </MultiGeometry>
 </Placemark>
 </Folder>
 <Style id="LineStyle00">
 <LabelStyle>
 <color>00000000</color>
 <scale>0.000000</scale>
 </LabelStyle>
 <LineStyle>
 <color>ff0055ff</color>
```

KML

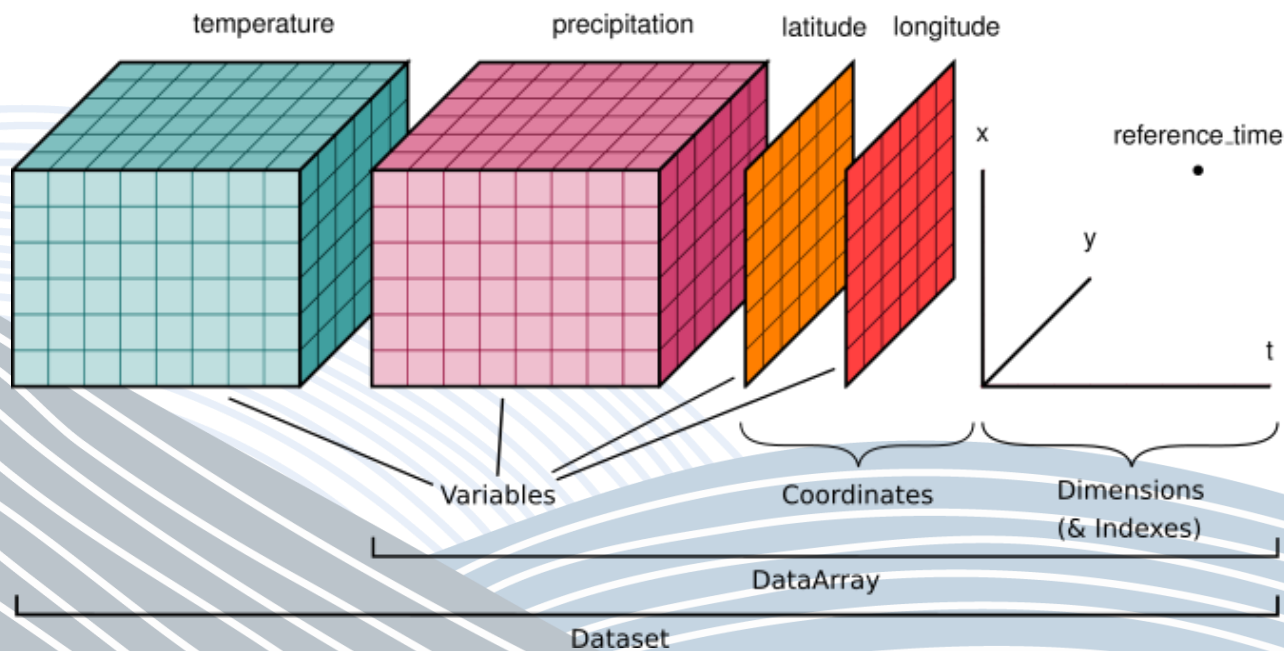


netCDF、GRIB

- 二進位，無法直接編輯
- 儲存多維度矩陣資料
- Python: netCDF4, xarray+cfgrib

```
<xarray.Dataset>
Dimensions: (latitude: 373, longitude: 363)
Coordinates:
 time datetime64[ns] 2023-07-11
 step timedelta64[ns] 00:00:00
 surface float64 0.0
 * latitude (latitude) float64 33.05 33.14 33.23 33.33 ... 67.73 67.83 67.92
 * longitude (longitude) float64 -13.99 -13.85 -13.71 ... 36.63 36.77 36.91
 valid_time datetime64[ns] ...
Data variables:
 t (latitude, longitude) float32 296.1 296.0 295.9 ... 283.0 282.9
Attributes:
 GRIB_edition: 2
 GRIB_centre: egr
 GRIB_centreDescription: U.K. Met Office - Exeter
 GRIB_subCentre: 5
 Conventions: CF-1.7
 institution: U.K. Met Office - Exeter
 history: 2025-07-01T12:15 GRIB to CDM+CF via cfgrib-0.9.1...
```

GRIB(左)、netCDF(下)資料表頭



```
root group (NETCDF4 data model, file format HDF5):
 Conventions: CF-1.6
 Metadata_Conventions: Unidata Dataset Discovery v1.0
 cdm_data_type: Grid
 comment: Sea Surface Height measured by Altimetry and derived variables
 contact: servicedesk.cmems@mercator-ocean.eu
 creator_email: servicedesk.cmems@mercator-ocean.eu
 creator_name: CMEMS - Sea Level Thematic Assembly Center
 creator_url: http://marine.copernicus.eu
 date_created: 2022-07-04T01:03:30Z
 date_issued: 2022-07-04T01:03:30Z
 date_modified: 2022-07-04T01:03:30Z
 geospatial_lat_max: 89.875
 geospatial_lat_min: -89.875
 geospatial_lat_resolution: 0.25
 geospatial_lat_units: degrees_north
 geospatial_lon_max: 179.875
 geospatial_lon_min: -179.875
 geospatial_lon_resolution: 0.25
 geospatial_lon_units: degrees_east
 geospatial_vertical_max: 0.0
 geospatial_vertical_min: 0.0
 geospatial_vertical_positive: down
 geospatial_vertical_resolution: point
 geospatial_vertical_units: m
 history: 2022-07-04 01:03:31Z: Creation
 institution: CLS, CNES
 keywords: Oceans > Ocean Topography > Sea Surface Height
 keywords_vocabulary: NetCDF COARDS Climate and Forecast Standard Names
 license: http://marine.copernicus.eu/web/27-service-commitments-and-licence.php
 platform: AltiKa, Cryosat-2 New Orbit, Haiyang-2B, Jason-3 Interleaved, Sentinel-3A, Sentinel-3B, Sentinel-6A
 processing_level: L4
 product_version: vDec2021
 project: COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)
 references: http://marine.copernicus.eu
 software_version: 19.2.0_DUACS_DT2021_baseline
 source: Altimetry measurements
 ssalto_duacs_comment: Sentinel-6A is the reference mission used for the altimeter inter-calibration processing
 standard_name_vocabulary: NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37
 summary: SSALTO/DUACS Near-Real-Time Level-4 sea surface height and derived variables measured by multi-satellite
 Global Ocean.
 time_coverage_duration: P1D
 time_coverage_end: 2022-07-01T12:00:00Z
 time_coverage_resolution: P1D
 time_coverage_start: 2022-06-30T12:00:00Z
 title: NRT merged all satellites Global Ocean Gridded SSALTO/DUACS Sea Surface Height L4 product and derived variables
 dimensions(sizes): time(1), latitude(720), longitude(1440), nv(2)
 variables(dimensions): int32 crs(), float32 time(time), float32 latitude(latitude), float32 lat_bnds(latitude, nv), float32 longitude(longitude), float32 lon_bnds(longitude, nv), int32 nv(nv), int32 sla(time, latitude, longitude), int32 err
```

<https://docs.xarray.dev/en/stable/user-guide/data-structures.html>

- 如.asc、GeoTIFF，或其他圖檔
- 一般TIFF需要.prj表明CRS和.tfw定位

X方向像素大小 (寬)  
 相對y軸旋轉  
 相對x軸旋轉  
 Y方向像素大小 (長)  
 左上角x座標  
 左上角y座標

## .asc網格檔案範例



```
1 PROJCS["TWD_1997_TM_Taiwan",GEOGCS["GCS_TWD_1997",DATUM["D_TWD_1997",SPHEROID["GRS_1980",
6378137.0,298.257222101]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433]],PROJ
ECTION["Transverse_Mercator"],PARAMETER["False_Easting",250000.0],PARAMETER["False_Northi
ng",0.0],PARAMETER["Central_Meridian",121.0],PARAMETER["Scale_Factor",0.9999],PARAMETER["
Latitude_Of_Origin",0.0],UNIT["Meter",1.0]]
```

.prj (WKT)

Export

OGC WKT

OGC WKT 2

ESRI WKT

PROJ.4

Proj4js

JSON

GeoServer

MapServer

Mapnik

PostGIS

Definition: JavaScript (Proj4js)

Open

Copy URL

Copy Text

Docs

Download

```
proj4.defs("EPSG:4326","+proj=longlat +datum=WGS84 +no_defs +type=crs");
```

epsg.io提供之格式

- 轉換NODASS下載之GDP軌跡為GeoJSON及KML
- [https://colab.research.google.com/github/ODBapp/2025\\_NODASS\\_workshop/blob/main/src/odb.ipynb](https://colab.research.google.com/github/ODBapp/2025_NODASS_workshop/blob/main/src/odb.ipynb)



# \*網路地圖服務



- Web Map Service (WMS)、Web Map Title Service (WMTS)、Web Feature Service (WFS)
- GetCapabilities
  - 回傳服務內容說明，例如圖層名、x/y/z格式、樣式、時間……等
- GetMap (WMS) / GetTile (WMTS) / GetFeature (WFS)
  - 回傳圖磚及圖徵(feature)
- \*GetFeatureInfo
  - Feature資訊 (eg. CMEMS)
- \*GetLegendGraphic
  - 圖例資訊

```
<Layer cascaded="0" fixedHeight="0" fixedWidth="0" opaque="0" queryable="0">
 <Name>EMAP</Name>
 <Title>臺灣通用電子地圖</Title>
 <Abstract>臺灣通用電子地圖</Abstract>
 <SRS>EPSG:3857</SRS>
 <SRS>EPSG:4326</SRS>
 <SRS>EPSG:3826</SRS>
 <SRS>EPSG:3825</SRS>
 <LatLonBoundingBox maxx="180.0" maxy="90.0" minx="0.0" miny="0.0" />
 <BoundingBox SRS="EPSG:3857" maxx="2.0037508342789248E7" maxy="2.3810769326496768E8" minx="0.0" miny="-7.081154551613623E-10" />
 <BoundingBox SRS="EPSG:3826" maxx="352136.448733" maxy="2807602.054796" minx="144847.277860" miny="2406316.107441" />
 <BoundingBox SRS="EPSG:3825" maxx="408747.911640" maxy="2992771.078492" minx="157736.290764" miny="2544566.280698" />
 <Style>
 <Name>default</Name>
 </Style>
</Layer>
<Layer cascaded="0" fixedHeight="0" fixedWidth="0" opaque="0" queryable="0">
 <Name>EMAP2</Name>
 <Title>臺灣通用電子地圖透明</Title>
 <Abstract>臺灣通用電子地圖透明</Abstract>
 <SRS>EPSG:3857</SRS>
 <SRS>EPSG:4326</SRS>
 <SRS>EPSG:3826</SRS>
 <SRS>EPSG:3825</SRS>
 <LatLonBoundingBox maxx="180.0" maxy="90.0" minx="0.0" miny="0.0" />
 <BoundingBox SRS="EPSG:3857" maxx="2.0037508342789248E7" maxy="2.3810769326496768E8" minx="0.0" miny="-7.081154551613623E-10" />
 <BoundingBox SRS="EPSG:3826" maxx="352136.448733" maxy="2807602.054796" minx="144847.277860" miny="2406316.107441" />
 <BoundingBox SRS="EPSG:3825" maxx="408747.911640" maxy="2992771.078492" minx="157736.290764" miny="2544566.280698" />
 <Style>
 <Name>default</Name>
 </Style>
</Layer>
```

```
<Layer queryable="1">
 <ows:Identifier>GLOBAL_ANALYSISFORECAST_PHY_001_024/cmems_mod_glo_phy_anfc_0.083deg_PT1H-m_202406/vo</ows:Identifier>
 <ows:Title>Northward surface velocity (vo)</ows:Title>
 <ows:WGS84BoundingBox>
 <ows:LowerCorner>-180 -80</ows:LowerCorner>
 <ows:UpperCorner>179.91668701171875 90</ows:UpperCorner>
 </ows:WGS84BoundingBox>
 <Style isDefault="true">
 <ows:Identifier>cmmap:balance</ows:Identifier>
 </Style>
 <Format>image/png</Format>
 <InfoFormat>application/json</InfoFormat>
 <InfoFormat>text/html</InfoFormat>
 <InfoFormat>text/xml</InfoFormat>
 <InfoFormat>application/vnd.ogc.gml/3.1.1</InfoFormat>
 <Dimension>
 <ows:Identifier>elevation</ows:Identifier>
 <UnitSymbol>m</UnitSymbol>
 <Default>-0.49402499198913574</Default>
 <Value>-0.49402499198913574</Value>
 </Dimension>
 <Dimension>
 <ows:Identifier>time</ows:Identifier>
 <ows:UOM>ISO8601</ows:UOM>
 <Default>2025-07-03T04:00:00.000Z</Default>
 <Value>2022-06-01T00:00:00Z/2025-07-11T23:00:00Z/PT1H</Value>
 </Dimension>
 <ows:Metadata>
 <ows:Title>WMTS Layer Metadata</ows:Title>
 <VariableInformation>
 <Id>vo</Id>
 <StandardName>northward_sea_water_velocity</StandardName>
 <Name>Northward surface velocity</Name>
 <Unit>m s-1</Unit>
 <MinimumValue>-1.5637902617454529</MinimumValue>
 <MaximumValue>1.5637902617454529</MaximumValue>
 <Colormap>balance</Colormap>
 <LogScale>false</LogScale>
 </VariableInformation>
 <DataCubeInformation>
 <admp_in_preparation>false</admp_in_preparation>
 <admp_released_date>2024-06-18T13:00:00.000Z</admp_released_date>
 <admp_updated>2025-07-02T06:09:25.281730Z</admp_updated>
 <admp_sample_period>hour</admp_sample_period>
 </DataCubeInformation>
 </ows:Metadata>
</Layer>
```

內政部WMS(左)及CMEMS WMTS(上) GetCapabilities範例

- RESTful
  - 以/分隔參數，順序有差別
  - [https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/MODIS\\_Terra\\_CorrectedReflectance\\_TrueColor/default/2012-07-09/GoogleMapsCompatible\\_Level9/{z}/{y}/{x}.jpg](https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/MODIS_Terra_CorrectedReflectance_TrueColor/default/2012-07-09/GoogleMapsCompatible_Level9/{z}/{y}/{x}.jpg)
- KVP (key-value pair)
  - 問號後以key=value形式加入參數，以&連接，大小寫可能有差別
  - 順序沒差別，可能會轉成HTML符號
  - WMTS
    - [https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/wmts.cgi?Service=WMTS&Request=GetTile&layer=MODIS\\_Terra\\_CorrectedReflectance\\_TrueColor&tilematrixset=GoogleMapsCompatible\\_Level9&style=default&TIME=2012-07-09&Version=1.0.0&TileMatrix={z}&TileCol={x}&TileRow={y}&Format=image%2Fjpeg](https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/wmts.cgi?Service=WMTS&Request=GetTile&layer=MODIS_Terra_CorrectedReflectance_TrueColor&tilematrixset=GoogleMapsCompatible_Level9&style=default&TIME=2012-07-09&Version=1.0.0&TileMatrix={z}&TileCol={x}&TileRow={y}&Format=image%2Fjpeg)
  - WMS
    - [https://gibs.earthdata.nasa.gov/wms/epsg3857/best/wms.cgi?time=2024-05-12T02:16:00Z&layers=GHR SST\\_L4\\_MUR\\_Sea\\_Surface\\_Temperature&transparent=true&format=image%2Fpng&service=WMS&request=GetMap&styles=&version=1.1.1&width=256&height=256&srs=EPSG%3A3857&crossOrigin=anonymous](https://gibs.earthdata.nasa.gov/wms/epsg3857/best/wms.cgi?time=2024-05-12T02:16:00Z&layers=GHR SST_L4_MUR_Sea_Surface_Temperature&transparent=true&format=image%2Fpng&service=WMS&request=GetMap&styles=&version=1.1.1&width=256&height=256&srs=EPSG%3A3857&crossOrigin=anonymous)

\*NMEA0183





## NMEA 0183

- 美國國家海洋協會制定之電子設備標準格式，通常為儀器轉至數位的最原始資料
- 只能使用ASCII符號
- \$ttsss,...,...\*hh

**\$**: 語句起始符號 **GP**: 辨識符號(talker identifier) **RMC**: 語句名(sentence identifier) **\***: 校驗和符號  
**2B**: 校驗和 (\$和\*之間XOR，HEX)

常見辨識符:

GP---: GPS

IN---: 整合導航 (eg., seapath)

HE---: 陀螺儀

WI---: 氣象(Weather Instruments)

常見語句名:

--GGA : GPS定位資訊 Global Positioning System Fix Data

--VTG : 地面速度資訊 Track Made Good and Ground Speed

--RMC : 推薦定位資訊 Recommended Minimum Specific GPS/TRANSIT Data

--ZDA : UTC日期和時間

--HDT : 真艏向 Heading, True

--DBT : 水深 Depth Below Transducer

--MWV : 風速風向 Wind Speed and Angle

--XDR : 傳感器 Transducer Measurement

```
$GPRMC,203522.00,A,5109.0262308,N,11401.8407342,W,0.004,133.4,130522,0.0,E,D*2B
```

\$--RMC,<1>,<2>,<3>,<4>,<5>,<6>,<7>,<8>,<9>,<10>,<11>,<12>\*hh

<1> UTC時間，hhmmss	<7> 對地速度(節)
<2> A=有效定位，V=無效定位	<8> 地面航向(000.0~359.9度，以真北為基準)
<3> 緯度ddmm.mmmm	<9> UTC日期，ddmmyy
<4> N或S (南北半球)	<10> 磁偏角 (000.0~180.0度)
<5> 經度dddmm.mmmm	<11> 磁偏角方向，E (東) 或W (西)
<6> E或W(東西半球)	<12> 模式指示 (A=自主定位，D=差分，E=估算，N=資料無效)